

**Nirma University
Institute of Technology**



Course Code :- 6CS103CC26
Course Name :- Operating System

Practical - 5

Student Name

Thakor Vinay

Student Roll No.

26MCA061

Faculty In Charge

Prof. Drashti Patel

Assignment Task 5: Employee Database Management

Objective

The purpose of this assignment is to develop a shell script that demonstrates basic database management operations using a text file in a Unix/Linux shell environment. The script is designed to perform six essential operations: inserting records, viewing records, searching records, sorting records, modifying records, and deleting records.

Implementation Details

The script begins by creating an `employees.txt` file if it does not already exist. The file stores employee information using pipe (|) as the field separator. The fields maintained are Employee ID, Employee Name, Designation, Department, Salary, and Age.

An interactive menu is implemented using a `while` loop and `case` control structure. It accepts the user's choice and performs the corresponding employee database operation.

The insert operation accepts employee details from the user and appends the new record to the `employees.txt` file using the `cat` command. The view operation displays all records stored in the file.

The search operation uses the `grep` command to find employee records based on Employee ID or Employee Name. The sorting operation separates the header from the records and sorts the employee records numerically according to Employee ID using the `sort` command.

The modify operation searches for an employee using the Employee ID, removes the existing record, and adds the updated employee information. The delete operation similarly searches for the specified Employee ID and removes the corresponding record from the file. Temporary files are used during modification, deletion, and sorting to safely update the database file.

Source Code

```
#!/bin/bash

if [ ! -f employees.txt ]; then
cat > employees.txt <<EOF
Emp_ID|Emp_Name|Designation|Department|Salary|Age
EOF
fi

while true
do
echo "
1. Insert Data
2. View All Records
3. Search Data
4. Sort File
5. Modify Data
6. Delete Record
7. Exit
```

```

"

read -p "Enter your choice: " ch

case $ch in

1)
read -p "Enter Employee ID: " id
read -p "Enter Employee Name: " name
read -p "Enter Designation: " designation
read -p "Enter Department: " department
read -p "Enter Salary: " salary
read -p "Enter Age: " age
cat >> employees.txt <<EOF
$id|$name|$designation|$department|$salary|$age
EOF
echo "Record inserted successfully."
;;

2)
cat employees.txt
;;

3)
read -p "Enter Employee ID or Name to search: " search
grep -i "$search" employees.txt
;;

4)
(head -n 1 employees.txt && tail -n +2 employees.txt | sort -t'|' -k1,1n) >
temp.txt
cat temp.txt > employees.txt
rm temp.txt
cat employees.txt
;;

5)
read -p "Enter Employee ID to modify: " id

record=$(grep "^$id|" employees.txt)

if [ -z "$record" ]; then
echo "Record not found."
else
read -p "Enter New Employee Name: " name
read -p "Enter New Designation: " designation
read -p "Enter New Department: " department
read -p "Enter New Salary: " salary

```

```

read -p "Enter New Age: " age

grep -v "^$id|" employees.txt > temp.txt

cat >> temp.txt <<EOF
$id|$name|$designation|$department|$salary|$age
EOF

cat temp.txt > employees.txt
rm temp.txt

echo "Record modified successfully."
fi
;;

6)
read -p "Enter Employee ID to delete: " id

if grep -q "^$id|" employees.txt
then
grep -v "^$id|" employees.txt > temp.txt
cat temp.txt > employees.txt
rm temp.txt
echo "Record deleted successfully."
else
echo "Record not found."
fi
;;

7)
exit
;;

*)
echo "Invalid choice."
;;

esac
done

```

Sample Data

The `employees.txt` file stores records in the following format:

Emp_ID	Emp_Name	Designation	Department	Salary	Age
101	Vinay	Manager	IT	50000	28
102	Amit	Developer	HR	40000	25
103	Neha	Analyst	Finance	45000	26

Output / Execution

The script provides an interactive menu through which the user can select the required database operation. The output displays the employee records after insertion, searching, sorting, modification, or deletion.

```
vinay@DevL0Q:/mnt/e/Codes/Nirma_SEM-1/OS/prac-5$ bash prac-5.sh
```

1. Insert Data
2. View All Records
3. Search Data
4. Sort File
5. Modify Data
6. Delete Record
7. Exit

```
Enter your choice: 1
Enter Employee ID: 101
Enter Employee Name: Vinay
Enter Designation: Manager
Enter Department: IT
Enter Salary: 700000
Enter Age: 25
Record inserted successfully.
```

1. Insert Data
2. View All Records
3. Search Data
4. Sort File
5. Modify Data
6. Delete Record
7. Exit

```
Enter your choice: 2
101|Vinay|Manager|IT|700000|25
```

1. Insert Data
2. View All Records
3. Search Data
4. Sort File
5. Modify Data
6. Delete Record
7. Exit

```
Enter your choice: 3
Enter Employee ID or Name to search: 101
101|Vinay|Manager|IT|700000|25
```

1. Insert Data
2. View All Records
3. Search Data
4. Sort File
5. Modify Data
6. Delete Record
7. Exit

Enter your choice: 4

Emp_ID|Emp_Name|Designation|Department|Salary|Age

101|Amit|Developer|HR|40000|25

102|Neha|Analyst|Finance|45000|26

103|Rahul|Manager|IT|50000|28

1. Insert Data
2. View All Records
3. Search Data
4. Sort File
5. Modify Data
6. Delete Record
7. Exit

Enter your choice: 5

Enter Employee ID to modify: 101

Enter New Employee Name: Vinay

Enter New Designation: Tech Lead

Enter New Department: IT

Enter New Salary: 800000

Enter New Age: 25

Record modified successfully.

Conclusion

This script demonstrates the implementation of basic DBMS operations using shell scripting and a text file. By combining commands such as `cat`, `grep`, `sort`, `head`, `tail`, and `rm` with `case` control structures and user input, the script provides an interactive employee record management system. The implementation reinforces fundamental concepts of file handling, searching, sorting, conditional branching, and record manipulation in a Unix/Linux shell environment.